

DEVELOPMENT OF AN ACTIVE SHOULDER PROSTHESIS WITH LOW-LEVEL CONTROL VALIDATION

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ABSTRACT

Proper functioning of the shoulder complex is imperative for upper limb motion. Therefore, amputees who undergo shoulder disarticulation or forequarter amputation require a prosthetic device which can replicate the functionalities of a human shoulder, in order to regain their upper limb motions. Different prosthetic devices are proposed for shoulder level amputees to assist motions of the shoulder complex. This paper proposed an externally powered prosthetic device for shoulder amputees. The shoulder prosthesis can perform shoulder flexion/extension, shoulder abduction/adduction, and shoulder internal/external rotation. Experiments are carried out with a low-level control method to validate the motion generation of the proposed prosthesis.

KEY WORDS

Assistive robots, Shoulder prosthesis, PID control

1 Introduction

1.5 out of every 1000 people in the world are subjected to amputations, and an estimated 10 million amputees are in the world [1]. An estimated 3%–15% has been subjected to amputations involving upper-limb [2]. There is a social responsibility to provide these people with the capability to carry out their activities of daily living (ADL). Upper-limb amputations are required due to various reasons, such as war casualties and accidents which result in trauma, congenital disorders, and diseases [3]. There are different stages of upper-limb amputations [4]: wrist disarticulation, transradial, elbow disarticulation, transhumeral, shoulder disarticulation and forequarter amputation. Out of them, the shoulder disarticulation and forequarter amputation result in complete loss of shoulder complex.

Prostheses are developed for amputees with various levels of amputation. However, development of shoulder prosthesis is mainly focusing on recreating the functions of shoulder complex, but most projects are limited to total arm replacement prostheses projects. Different prosthesis options are available for shoulder amputees. There are active and passive prostheses, with the latter primarily serving for cosmetic purposes [5]. Passive prostheses offer minimal support for the amputee to carry out their

work. They are comparatively popular among amputees owing to the wide availability compared with active prostheses, lower cost, and reliability of the control mechanism. However, significant limitations can be seen in maneuvering and replicating motions [6]. Therefore passive shoulder prostheses are inadequate to enable the amputee with the motion capabilities of the human shoulder complex. Active shoulder prostheses can replicate human shoulder complex motions to a greater extent to reduce the impact of shoulder loss. Some state-funded prosthesis development projects across the world have managed to achieve the motions through externally powered shoulder prostheses [7]. Externally powered active prostheses incur a large development cost and hence are expensive, which reduce their accessibility for all amputees. Therefore, it is necessary to carry out further research in order to develop low-cost commercialized active prostheses. Active shoulder prostheses can be body powered (cable operated), externally powered (usually myoelectric-controlled) or hybrid type [8]. Electromyography (EMG) signals based control methods have been used by researchers as a high-level control method with its real-time controllability and reliability over other methods [12]–[14]. For a prosthetic device to be accepted for consistent use by an amputee, several factors such as functionality, aesthetic appearance, ease of use, and low weight are deemed decisive [9].

This paper is proposed an externally powered prosthesis for shoulder amputees to replicate the motion of the shoulder complex, and enable the use of a modular transhumeral prosthetic device, to carry out the desired motions. During the design phase, a mechanical shoulder complex with 3 degrees of motion (DOF) (abduction/adduction, flexion/extension, and internal rotation/external (humeral) rotation (Figure 1)) was designed and developed. Custom designed gear assemblies with brushless DC motors (BLDC) were used to drive each DOF of the prosthesis. In order to drive the respective motors, a low-level closed loop control method based proportional-integral-derivative (PID) is used. Experiments are carried out to validate the motions of the developed prosthesis and the effectiveness of the low-level control method is tested.

The arrangement of this paper is as follows: Section 2 presents shoulder prostheses developments to date. Sec-

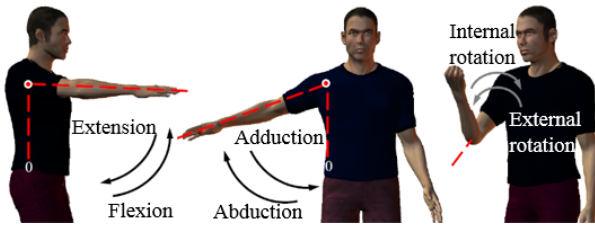


Figure 1: Three degrees of freedom of shoulder: a) flexion/extension b) abduction/adduction c) internal rotation/external rotation (humeral rotation) [4]

tion 3 consists of mechanical design and development details related to the shoulder prosthesis. Section 4 describes the implemented lower-level control method for prosthesis motion validation. In Section 5 conducted experiments and results are discussed and the conclusion is presented in Section 6.

2 Related Work

Active prosthetic devices have been manufactured for the use of upper-limb amputees, on a commercial scale, by manufacturers such as Ottobock [10], Utah arm [11] and Deka arm [12]. However, active upper-limb prostheses for transhumeral and transradial amputees are not commonly available than that of hand prostheses [5]. Furthermore, the active prosthetic devices for shoulder level amputees are still largely at the research level. Notable developments have been done for shoulder prostheses including the modular prosthetic limb by the Advanced Physics Laboratory of Johns Hopkins University (JHU APL). Shoulder level of JHU prosthesis consists 3DOF shoulder which controls based on EMG signals and it can replicate the user desired motions [13]. Utah arm consists with 2 DOF lockable shoulder. It can be used for amputees with shoulder disarticulation or forequarter amputation. Utah arm has fabricated with injection-molded plastics in order to minimized the weight and cost [11]. DEKA Integrated Solutions Corporation has built a prosthetic arm which could be used by amputees with different levels of amputation—for transradial, transhumeral and shoulder amputees. DEKA arm also offers an active prosthetic device for shoulder amputees (of very short residual limb transhumeral, shoulder disarticulation and interscapulothoracic (forequarter) levels of amputation). DEKA arm consists of 10 DOF including 3DOF shoulder prosthesis [12]. Furthermore, several researches have been conducted to identify an active prosthesis solution for shoulder amputees. Gruppioni et al. [14] have conducted research on the development of a prosthetic limb for shoulder level amputees. Cantilever joint had been used to support the prostheses along with two powered DOF. Troncosi et al. have developed and clinically tested a 2DOF shoulder articulation device, which is designed to be used by proximal transhumeral amputees as well as interscapulothoracic (forequarter). Even though there are few shoulder prostheses projects as mentioned, most of them are yet to be perfected. Therefore further research on shoulder prostheses are to be carried out.

Table 1: Specifications of motor

Description	Specifications
Voltage	24 V
Weight	110 g
Maximum continuous torque	83.4 mNm
Gear-head model	45 mm, 2.0 Nm, Maxon
Gear-head weight	250 g
Reduction ratio	89 : 1

3 Mechanical Design

The primary design parameters to be appropriately considered when developing an externally powered prostheses are the torque required at the joints, the angular velocity of the joint, and the movable ranges. Typical human shoulder generates average torques of 16.9 Nm, 16.9 Nm and 8.46 Nm for flexion/extension, abduction/adduction, and internal rotation respectively [15]. In addition, the rotational speed of a typical human shoulder is in the range of 4 to 10 rpm (0.41–1.05 rad/sec) [15]–[17]. The flexion/extension of healthy human ranges from (–30) to 180 deg. Abduction/adduction ranges from 0 to 150 degrees and internal rotation ranges from (–90) to 90 degrees. These parameters should be taken into account when designing the prosthesis.

On the other hand, the transhumeral prosthesis which should be connected with the shoulder prosthesis is expected to have approximate weight of 5 kg [15]–[17]. Moreover, it is necessary to design the prosthesis so that it should be able to carry a payload at the terminal device/hand. Therefore, based on these requirements, a preliminary calculation is carried out to identify the basic torque, speed and movable range requirements.

The weight of the prosthesis should be kept to a minimum as amputees have to wear it for a longer period of time. Multiple surveys have been conducted to determine the factors affecting amputees' response over extended use of prosthesis, and these have clearly highlighted the need for lightweight prostheses [18]. The device, along with the three joints and connectors has to be compact and smaller in size to make the amputee feel the minimal difference between the human limb and the prosthesis. The requirement of appropriate torque, speed, weight, and size can only be obtained by BLDC motors because BLDC motors have high torque/weight ratio. Therefore brushless motor (EC 45–4 pole, Maxon) was selected to fulfill the requirement. Table 1 shows the specifications of the motor.

External gear assembly is assembled with the gear ratio 1:12 to the maximum torque of 2 Nm of brushless DC motor, in order to obtain the proposed required torque of 24 Nm for flexion/extension and abduction/adduction (Figure 2). All the external gearboxes consist of constant mesh spur gears. 3:1 and 4:1 gear reduction is done with two sets of spur gear pairs. Abduction/adduction motor is rotating relative to the flexion/extension motor. Internal/external rotation motor operates relative to the abduction/adduction motor (Figure 3).

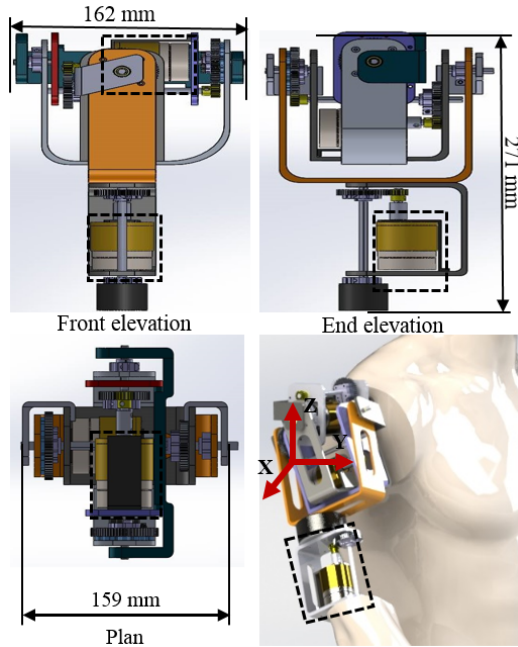


Figure 2: Main components of shoulder prosthesis

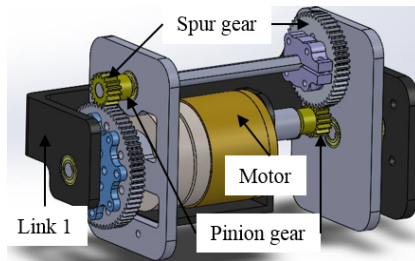


Figure 3: 3D CAD model of the gearbox assembly

Two motors of flexion/extension and abduction/adduction are aligned perpendicular to each other and one below the other (Figure 2). The 3rd degree of freedom humeral rotation motor is attached to the upper arm (Figure 4). In the design, abduction/adduction motor and humeral rotation motor operate relative to the flexion/extension motor. In order to eliminate singular point from flexion/extension and abduction/adduction, motion limits are implemented in the design and they function as safety mechanisms to protect the user if the device malfunctions. The humeral rotator is fabricated as a modular unit (Figure 4). Therefore, modular unit can be used by an amputee with a short humeral stump, who cannot compensate for the humeral rotation by own self (Figure 5).

All the mechanicals linkages and connectors were fabricated using Aluminium (6061). The link-1 in Figure 3 were fabricated by sand casting and machined to get the necessary surface.

4 Low-level control method for validation of the motion of the shoulder prosthesis

The control algorithm is designed to control synchronized motions along all three axes individually and simultane-

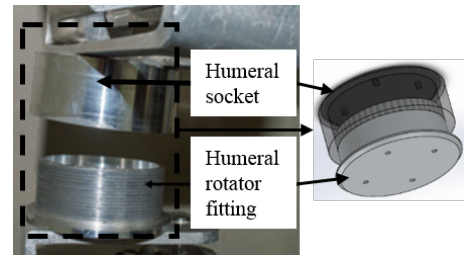


Figure 4: Shoulder prosthesis main components

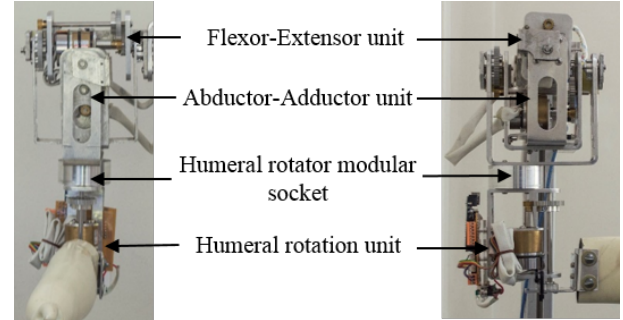


Figure 5: Developed shoulder prosthesis

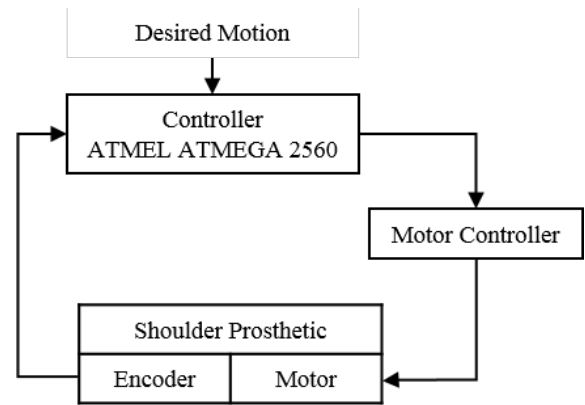


Figure 6: Architecture of hardware control system

ously. The preliminary control method is aimed at obtaining controlled motion in the three axes for the abduction adduction, flexion extension and internal rotation external rotation. Three OEM BLDC motor drivers (DEC 50W/5A EC, Maxon) are used to drive the motors. The motors used for the prosthesis does not utilize any encoder for position control (Figure 6). Therefore, in order to control the prosthesis as required, the signal from the hall sensors of the BLDC motor is utilized and it is provided by the motor driver module as an output. Using this output, the actual or current angular velocity and the position of the prosthesis is calculated.

The lower-level controlling is based on a simple proportional-integral-derivative (PID) controlling method. The structure of the algorithm is shown in Figure 7. The user can provide the desired input angle (determined by any high-level control scheme such as electromyography (EMG) or electroencephalogram (EEG) signal based control methods) and that angle is compared with the current position of the BLDC motor. Depending on the error, addi-

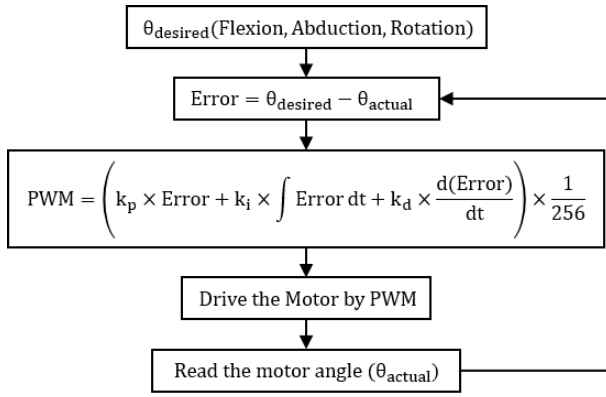


Figure 7: Basic control algorithm

Table 2: Movable range of human shoulder and developed shoulder prosthesis

Criterion	Human Shoulder	Developed Prosthesis
Degrees of Freedom	3	3
Motion ranges (deg.)		
Flexion-Extension	180 : (−30)	170 : (−40)
Abduction-Adduction	150 : 0	208 : (−12)
Internal-Ext. rotation	90 : (−90)	90 : (−90)
Lateral distance bet. axes	0	14.7 mm
Average Torque (Nm)		
Flexion / extension	18	24
Abduction / adduction	18	24
Internal / external rotation	9	6
Average velocity	4–10 rpm	5 rpm

tional angle needed to be moved for obtaining the desired location will be fed to the motor as a PWM signal. The PID coefficients are tuned by trial and error (Err) method.

$$PWM = \left[k_p \times Err \right] + \left[k_i \times \int (Err) dt \right] + \left[k_d \times \frac{d(Err)}{dt} \right]$$

5 Experiments and Results

Experiments were performed in order to validate the effectiveness of desired motion generation capability of the shoulder prosthesis. In the first experiment, desired angle signal was input along each axis. As the desired input signal, a sine wave was selected due to the capability of generating repetitive signal along positive and negative motions. The angle of the prosthesis was measured throughout the period of the experiment. Shoulder prosthesis was closely followed the sine wave signal. The results are shown in Figure 8.

The second experiment was conducted to validate the effectiveness of real-time motion following of the shoulder prosthesis. The experimental path is shown in Figure 10. The experiment setup was shown in Figure 11. The experiment was carried out by fixing an IMU to a male subject (26 years old) and obtaining inputs based on the subjects shoulder movement. IMU (MPU-6050 6DOF gyro

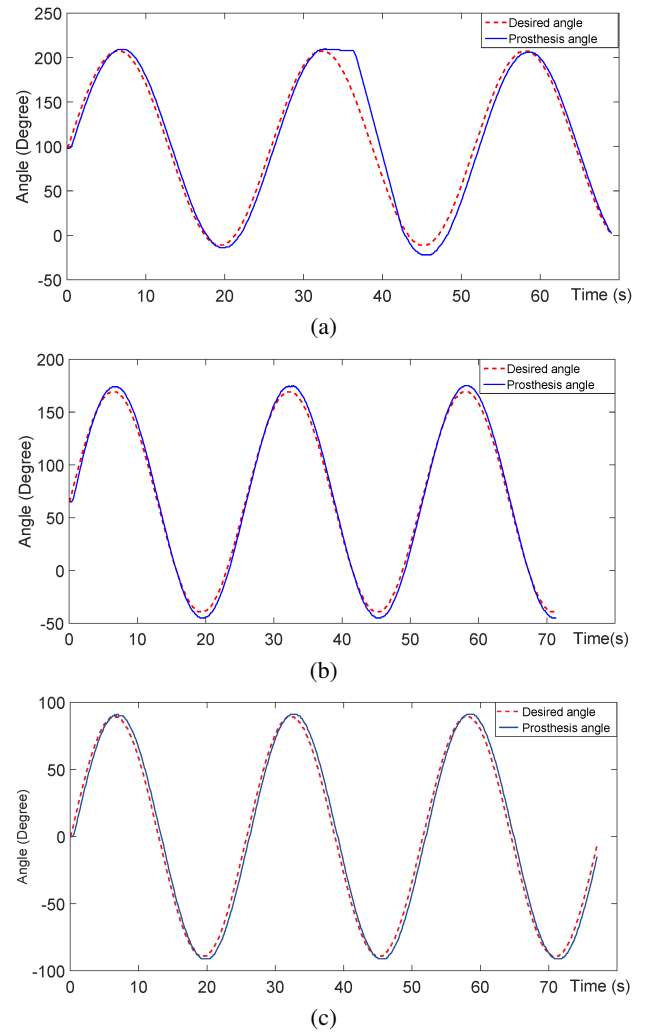


Figure 8: Shoulder prosthesis (a) abduction / adduction, (b) flexion / extension, and (c) humeral rotation motions plotted against the input signal

accelerometer) with Kalman filtered data acquisition algorithm was used to obtain accurate angle measurements. The subject was asked to move his upper-limb holding an electric drill in hand, in a rectangular path along a frontal (coronal) plane. The initialization of the shoulder prosthesis was done by setting it to the position depicted in Figure 10 and resetting all the encoders and Inertial Measurement Unit (IMU) values to zero. The obtained results are shown in Figure 12. In order to clearly illustrate the two motions separately, flexion/extension angles were taken as minus angles. The shoulder prosthesis closely followed the desired angle provide by the IMU throughout the experiment.

The developed shoulder prosthesis was able to generate proposed motion ranges similar to a human shoulder. Table 2 validates the effectiveness of shoulder prosthesis motion range capability (Figure 9). However, the lateral distance between flexion/extension axis and abduction/adduction axis in shoulder prosthesis was 14.7 mm, whereas the actual distance in a human shoulder is zero. The movable range was restricted by the mechanical limits in the design.

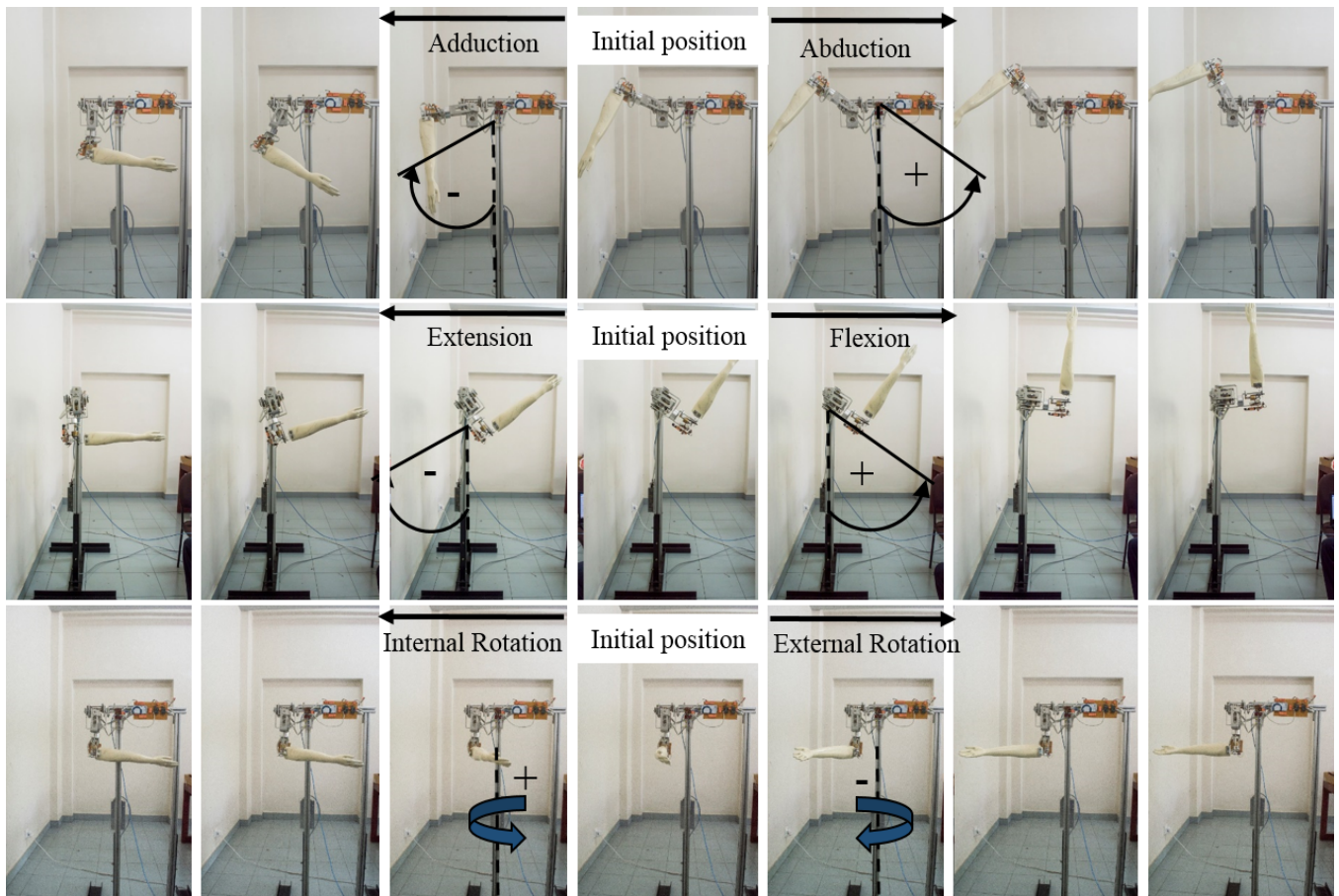
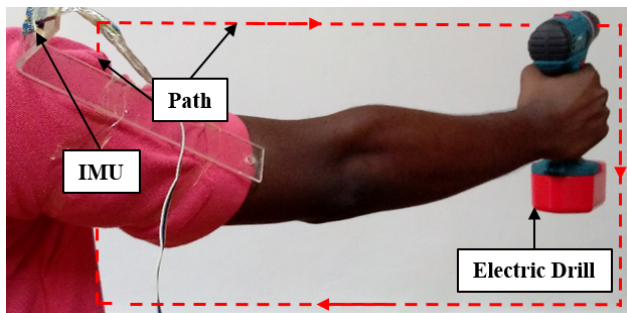
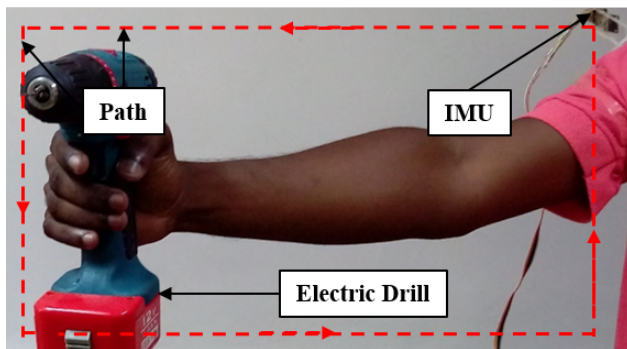


Figure 9: Motions of the shoulder prosthesis



(a)



(b)

Figure 10: Selected daily activity for experimentation

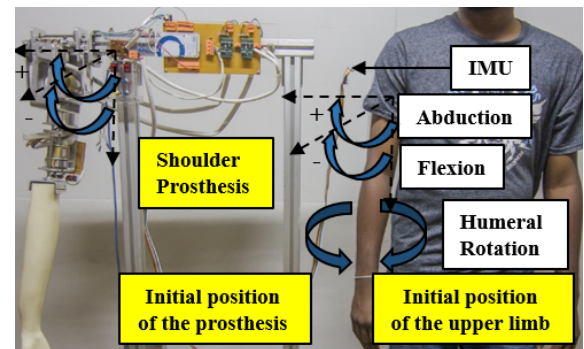


Figure 11: Experimental setup

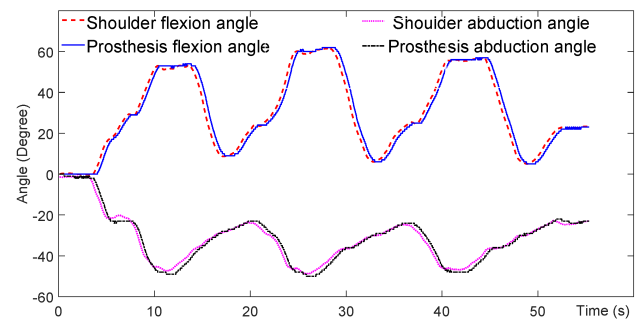


Figure 12: Validation of prosthesis motion generation for a selected daily living activity

6 Conclusion

In this study, a novel 3 degrees of freedom externally powered shoulder prosthesis was proposed. The proposed shoulder prosthesis replicates the motions of shoulder flexion/extension, abduction/adduction, and internal rotation/external rotation. The lateral distance between flexion/extension axis and abduction/adduction axis in a healthy human shoulder is approximately zero. Due to the significant distance between lateral axes, users may feel uncomfortable. Congenital absence and lack of stability cause such discomfort [19], [20]. Therefore, it is important to reduce this distance as much as possible. In the proposed design, this requirement was considered and the lateral distance between flexion/extension axis and abduction/adduction axis was reduced to 14.7 mm. Moreover, the shoulder prosthesis was able to fulfill the required motion ranges similar to the human shoulder. The effectiveness of the shoulder prosthesis and the lower-level control method was evaluated using experiments. The results of the experiments showed that the developed shoulder prosthesis closely followed the given input signals.

The total weight of the developed shoulder prosthesis is 3.2 kg. In the future works, the design can be optimized further by the adoption of lighter motors which will significantly help to reduce the weight of the prosthesis. Moreover, the future work should incorporate the development of shoulder socket to attach the developed prosthesis to the amputee's body. On the other hand, in terms of the controlling aspect of the prosthesis, the low-level control system needs to be improved further by validating with multiple users. Moreover, integration of the low-level control system with a high-level control method, such as EMG or EEG based control, should be studied in the future work.

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